

Easy (10 questions)

1.

What is static routing?

- A) Routes learned dynamically
- B) Routes manually configured
- C) Routes learned from neighbors
- D) Routes discovered by DNS

Answer: B

2.

Which routing protocol uses hop count as its metric?

- A) OSPF
- B) EIGRP
- C) RIP
- D) BGP

Answer: C

3.

Which routing protocol is Cisco proprietary?

- A) RIP
- B) OSPF
- C) EIGRP
- D) BGP

Answer: C

4.

What type of routing protocol is OSPF?

- A) Distance Vector

- B) Link-State
- C) Hybrid
- D) Path Vector

Answer: B

5.

Which routing method requires manual configuration?

- A) Static routing
- B) Dynamic routing
- C) Hybrid routing
- D) None

Answer: A

6.

What command is used to configure a static route on a Cisco router?

- A) ip route
- B) route add
- C) set route
- D) configure route

Answer: A

7.

Which routing protocol is suitable for very large networks like the Internet?

- A) RIP
- B) OSPF
- C) BGP
- D) IGRP

Answer: C

8.

Dynamic routing protocols automatically adjust routes based on:

- A) Manual input
- B) Network topology changes
- C) User commands
- D) DNS records

Answer: B

9.

What is the maximum hop count for RIP?

- A) 10
- B) 15
- C) 20
- D) 30

Answer: B

10.

Which routing protocol floods Link State Advertisements?

- A) RIP
- B) EIGRP
- C) OSPF
- D) BGP

Answer: C

Medium (30 questions)

11.

Which routing protocol combines distance vector and link-state features?

- A) RIP
- B) OSPF
- C) EIGRP
- D) BGP

Answer: C

12.

What happens if a static route's next-hop IP is unreachable?

- A) Router recalculates route
- B) Route is dropped until manually fixed
- C) Router sends ICMP redirect
- D) Router uses dynamic route automatically

Answer: B

13.

Which protocol uses Autonomous System (AS) numbers?

- A) OSPF
- B) RIP
- C) BGP
- D) EIGRP

Answer: C

14.

How often does RIP send routing updates by default?

- A) Every 5 seconds
- B) Every 10 seconds
- C) Every 30 seconds

D) Every 60 seconds

Answer: C

15.

Which protocol uses the DUAL algorithm?

A) OSPF

B) EIGRP

C) RIP

D) BGP

Answer: B

16.

Which protocol is classless and supports VLSM?

A) RIP version 1

B) RIP version 2

C) Static routing

D) IGRP

Answer: B

17.

What is the metric used by OSPF to select the best route?

A) Hop count

B) Bandwidth (cost)

C) Delay

D) AS path length

Answer: B

18.

Which routing protocol does NOT send periodic full routing table updates?

- A) RIP
- B) OSPF
- C) IGRP
- D) BGP

Answer: B

19.

Which of the following is NOT true about static routes?

- A) They require manual configuration
- B) They consume router CPU constantly
- C) They do not change unless configured
- D) They have no routing overhead

Answer: B

20.

What type of routing protocol is BGP?

- A) Distance Vector
- B) Path Vector
- C) Link-State
- D) Hybrid

Answer: B

21.

Which command can you use to verify the static routes on a router?

- A) show static routes
- B) show ip route static
- C) show ip route
- D) show route

Answer: C

22.

What is a disadvantage of static routing?

- A) Uses more bandwidth
- B) Requires high CPU
- C) Does not scale well
- D) Automatically adapts to topology

Answer: C

23.

How do dynamic routing protocols prevent routing loops?

- A) By manually configuring routes
- B) By periodic flooding of updates
- C) Using algorithms like split horizon, route poisoning, hold-down timers
- D) By ignoring bad routes

Answer: C

24.

Which routing protocol is considered an Exterior Gateway Protocol (EGP)?

- A) OSPF
- B) RIP
- C) BGP
- D) EIGRP

Answer: C

25.

What is the default administrative distance of static routes?

- A) 90
- B) 110
- C) 120

D) 1

Answer: D

26.

Which protocol uses multicast address 224.0.0.5 to send updates?

A) RIP

B) OSPF

C) BGP

D) EIGRP

Answer: B

27.

Which routing protocol supports route summarization?

A) RIP v1

B) OSPF

C) RIP v2

D) Both B and C

Answer: D

28.

In OSPF, the area used to connect multiple areas is called:

A) Backbone Area 0

B) Stub Area

C) Virtual Link

D) Non-backbone Area

Answer: A

29.

Which metric is NOT used by EIGRP?

- A) Bandwidth
- B) Delay
- C) Hop count
- D) Load

Answer: C

30.

Which command displays the routing protocols running on a router?

- A) show protocols
- B) show ip protocols
- C) show routing
- D) show ip route

Answer: B

Hard (10 questions)

31.

How does EIGRP achieve fast convergence?

- A) By flooding the entire routing table every 30 seconds
- B) By using the Diffusing Update Algorithm (DUAL) to calculate backup routes
- C) By using hop count as a metric
- D) By using periodic hello packets only

Answer: B

32.

In OSPF, what is the function of a Designated Router (DR)?

- A) Forward routing updates between routers on a multi-access network
- B) Backup router for all routers
- C) Perform routing only for backbone area
- D) Aggregate routes from different areas

Answer: A

33.

What does the administrative distance represent?

- A) The speed of the route
- B) The reliability of the router interface
- C) The trustworthiness of a routing protocol source
- D) The number of hops to the destination

Answer: C

34.

Which command will show you the OSPF neighbor relationships?

- A) show ospf interface
- B) show ip ospf neighbor
- C) show ospf status
- D) show ip route ospf

Answer: B

35.

What is the metric calculation formula for EIGRP?

- A) Hop count only
- B) Bandwidth + Delay (plus optional load and reliability)
- C) Cost based on bandwidth only
- D) AS Path length

Answer: B

36.

Which routing protocol does NOT support authentication natively?

- A) RIP v1
- B) OSPF
- C) EIGRP
- D) BGP

Answer: A

37.

How does RIP prevent routing loops?

- A) Split horizon, route poisoning, and hold-down timers
- B) DUAL algorithm
- C) Link State Advertisements
- D) AS Path filtering

Answer: A

38.

What is the default administrative distance of OSPF?

- A) 90
- B) 110
- C) 120
- D) 1

Answer: B

39.

Which of the following is true about BGP?

- A) It is an Interior Gateway Protocol
- B) Uses distance vector algorithm only
- C) Uses TCP port 179 for communication

D) Uses hop count as its primary metric

Answer: C

40.

What command would you use to configure a default static route pointing to next-hop 192.168.1.1?

A) ip route 0.0.0.0 0.0.0.0 192.168.1.1

B) default route 192.168.1.1

C) ip default-gateway 192.168.1.1

D) route add default 192.168.1.1

Answer: A